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PLC Programming Instructions

Programmable Logic Controllers

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Agenda

Basic PLC Programming Instructions

PLC Timers

PLC Counters

Advanced PLC Programming Instructions

Data Manipulation Instructions

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Basic PLC Programming Instructions

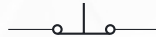
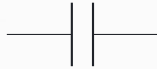
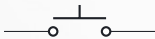
The 3 Basic Instructions

- Ladder logic doesn't care what the input *is*
- All that matters is on/off
- Remember, these were intentionally designed to look like relay contacts and coils



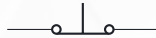
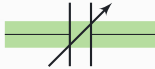
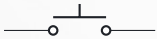
Examine if Closed (XIC)

True when the input is true (on)



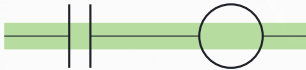
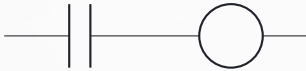
Examine if Open (XIO)

True when the input is false (off)



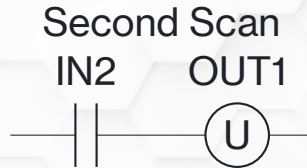
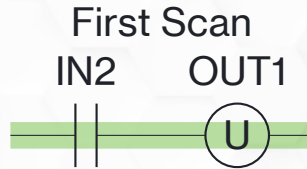
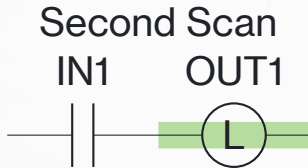
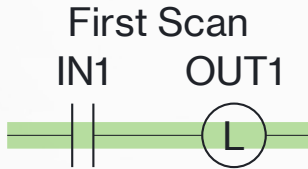
Output Energize (OTE)

True when the logic before it is true



Output Latch (OTL) and Unlatch (OTU)

Once turned on the outputs *stays on* until explicitly unlatched



Take Caution with Latches

- Be *very* careful with latches—my recommendation is to not use them as much as possible
- OTE is *volatile*—when power cycles the output is set to 0 (off)
- OTL is *NOT* volatile—if it's on when power goes out it will be on when power is restored
- This can and has caused unwanted/unsafe machine motion

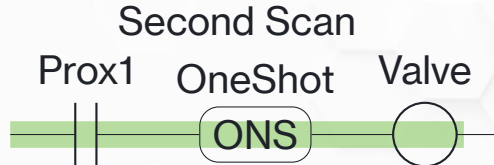
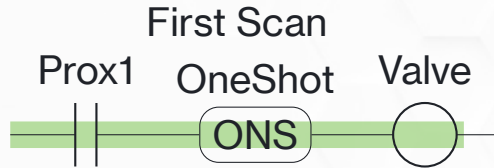
Latch/Unlatch on Key Platforms

How Key Platforms Handle Latch/Unlatch on Power Cycle

Manufacturer	Equivalent Instructions	Power Cycle Behavior	Memory Control Mechanism
Allen-Bradley (Logix5000)	OTL/OTU	Retains state across power cycles (non-volatile)	Tied directly to the instruction choice (OTL/OTU memory retention vs. OTE non-retention)
Siemens (TIA Portal)	S (Set)/ R (Reset)	Resets to 0 unless declared as Retentive	Bit/DB retain settings in tag table
Beckhoff (TwinCAT)/ CODESYS	S (Set)/ R (Reset)	Resets to 0 unless located in Retain memory	RETAIN or PERSISTENT keywords
Mitsubishi (GX Works)	SET/RST	Retains state if mapped to Latch Range	Latch (L) device memory range
Omron (Sysmac Studio)	Set/Rst	Resets to 0 unless declared as Retentive	Retain attribute checkbox on VAR

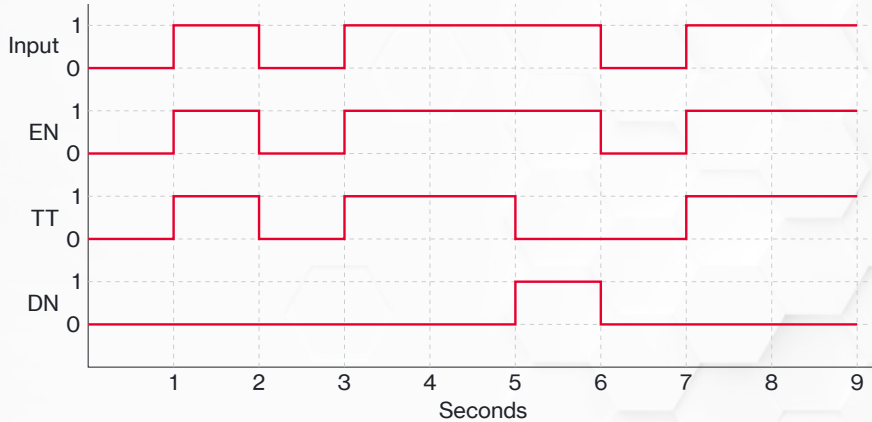
Oneshot

A oneshot is true for *one scan* while the logic before it is true—the logic has to turn false to reset it

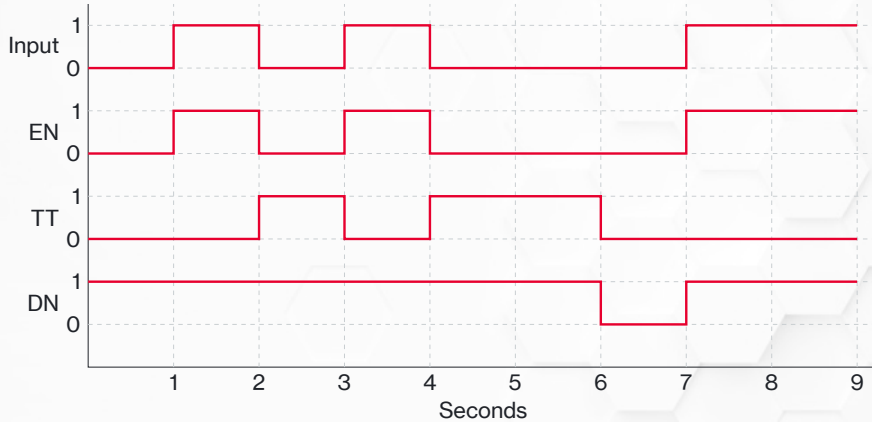


PLC Timers

TON Timing Diagram

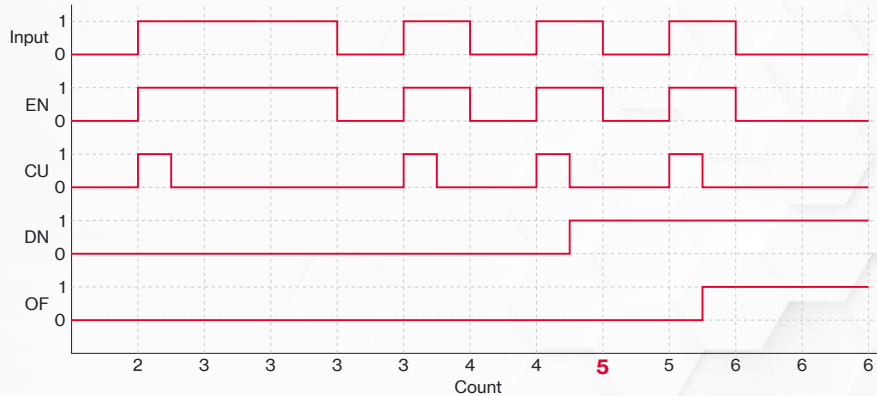


TOF Timing Diagram



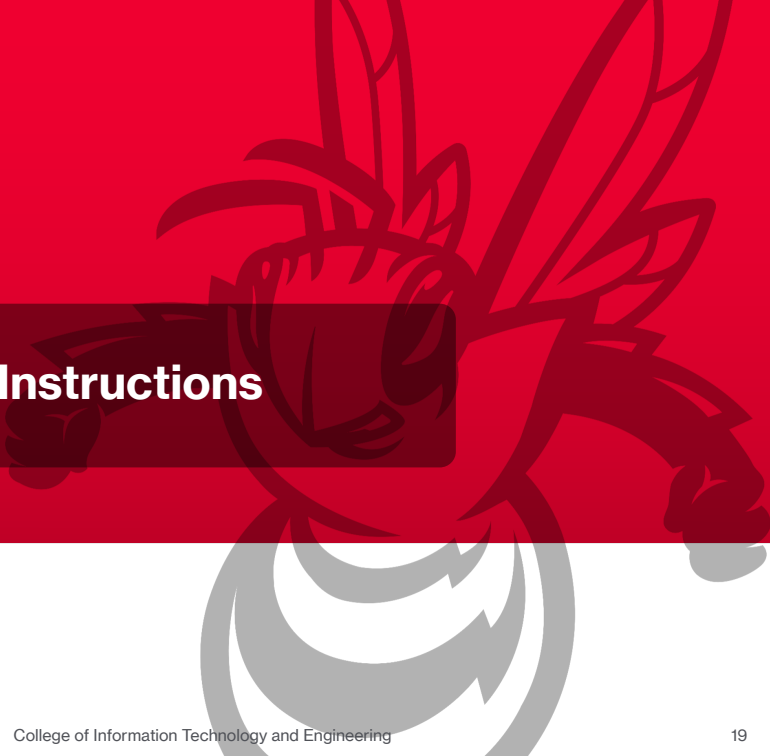
PLC Counters

CTU Timing Diagram



Advanced PLC Programming Instructions

Data Manipulation Instructions



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